

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0028, 2026-0806)

#	Quantity	Done	Remaining	Of
1	Definitions to define	11	2 (the three powerdomains, D^∞)	≈ 13
2	Numbered results complete	15 (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8, Lem 10, Thm 11, Prop 15, Lem 17, Lem 19, Lem 20, Thm 22, Lem 23)	14	29
2a	— of which partially proved	Thm 16 (the algebraic-lattice conjunct proved; the $Fp(D) \hookrightarrow (D \rightarrow D)$ embedding conjunct not stated)	—	—
3	Unnumbered prose claims proved	12	—	—
4	Mathlib foundations reused	12	—	12
5	Theorems in the development	384 live (+6 commented out as unused)	—	—
6	sorry in the development	—	1 : thm18 in Skeleton/Section6.lean	—

Round r0028 ran five agents in parallel and roughly doubled the development: $27 \rightarrow 33$ modules, $4440 \rightarrow 8212$ lines, $199 \rightarrow 384$ theorems, $9 \rightarrow 15$ numbered results. Every new result was kernel-audited with `#print axioms`: all depend only on `propext`, `Classical.choice`, `Quot.sound`, and none on `sorryAx`.

#	r0028 stream	Landed
1	CoalescedSum.lean, Skeleton/Sum.lean	$D + E$ as a cpo (<code>sumSup</code> , <code>sumCpo</code>), then <code>lem10_sum</code> , <code>lem17_sum</code> , <code>lem17_smash</code> — Lemma 10 at 6 of 6 conjuncts, Lemma 17 at 5 of 5
2	IdealCompletion.lean	Theorem 11 , both halves, on Mathlib’s <code>Order.Ideal</code> ; §5 unblocked
3	FinitaryProjectionPoset.lean, Skeleton/Section6b.lean	$Fp(D)$ and $Fc(D)$ as posets, Theorem 16 (algebraic-lattice conjunct), Lemma 20 , and <code>IsClosure.domain_range</code> — Lemma 19 at the paper’s strength

#	r0028 stream	Landed
4	MinimalUpperBounds.lean	minimal upper bounds, U , U^∞ , and <code>isPlotkinOrder_iff_mubClosure</code> — a characterization the paper does not state. Theorem 18 still open
5	UniversalDomain.lean	Theorem 22 and Lemma 23 , with <i>representable</i> defined from §7; opens the route to D^∞

A name clash lake build could not catch. Streams 3 and 5 each defined `IsClosure.apply_sSup_of_directed` and `isClosure_sSup`. The build passed at 971 jobs because no module imported both; the clash appeared the moment anything did — here, an axiom audit importing the pair. `isClosure_sSup` was the *same* statement proved twice; the single copy now lives in `Skeleton/Section6.lean` beside `IsClosure`, which both modules already import, so no call site changed. The two `apply_sSup_of_directeds` were *different* statements sharing a name — one indexed by a set of the subtype $\mathfrak{r}(\text{im } r)$, one by an ambient $D : \text{Set } \alpha$ with $D \subseteq \text{im } r$ — so the subtype form was renamed `IsClosure.apply_sSup_val_image_of_directed`. **A green build is not evidence that parallel work composes;** importing every new module together is.

Row 5 counts lines matching `^(@[...] )?(theorem|lemma)` across the 27 modules.

The sorry burn-down (from r0026). The *sorry*s are deliberate scaffolding: fixed *statements* of outstanding results, confined to `ScottDomains/Skeleton/`, one file per agent worktree, so that three agents can prove them in parallel without any agent editing a declaration another depends on. Every other module remains *sorry*-free, and the count above is the burn-down metric — it goes $10 \rightarrow 0$. Round r0027, run as three agents in parallel, took it **$10 \rightarrow 1$** .

#	Open statement	Result	State after r0027
1	prop15	Prop 15 — every bounded complete domain is bifinite	proved
2	thm18	Thm 18 — $D, D \rightarrow D$ domains $\implies D$ bifinite	sorry — the paper gives no proof, citing Smyth [Smy83a]; the obstacle is recorded in the docstring
3	lem19	Lem 19 — the image of a closure is a domain	proved , via <code>IsClosure.rangeCompletePartialOrder</code>
4–7	lem10_prod, lem10_smash, lem10_lift, lem10_strict	Lem 10 — bounded completeness closed under $\times, \otimes, ()^\perp, \rightarrow^\perp$. The $\rightarrow$ conjunct is already proved (Thm 7’s bounded-complete half, r0007)	all four proved ; Lem 10 is then 5 of 6 conjuncts — $D + E$ is not stated
8–10	lem17_prod, lem17_lift, lem17_fun	Lem 17 — bifiniteness closed under $\times, ()^\perp, \rightarrow$	all three proved ; Lem 17 is then 3 of 5 conjuncts — $D \otimes E$ and $D + E$ are not stated

The + conjuncts, closed in r0028. `CoalescedSum.lean` was 181 lines of ingredients with no `sSup` and no `cpo` instance, so there was no $D + E$ to state a conjunct over. It now has both, and the guard is the membership condition `IsNonBotSum`

(`sumCandidate (sumBase s)`) — the defining predicate of the subtype — not directedness, so the defect fixed in `ScottHom` and then in `Smash` did not recur a third time. One wrinkle the smash did not have: $\alpha \otimes \beta$ carries no `SupSet`, so a summand must be selected first; that selection is not a second guard, because a set with an upper bound at all lies in one summand.

The smashSup defect (r0027). `lem10_smash` was not merely open: as `smashSup` stood, it was **false**, and the kernel confirmed a refutation. `smashSup` branched its dte on the base being nonempty *and directed*, so a merely **bounded** non-directed base fell to the adjoined $\perp$, which is not even an upper bound. Witness: $D = \text{Prop} \times \text{Prop}$, $E = \text{Prop}$, $s = \{\uparrow((\text{True}, \text{False}), \text{True}), \uparrow((\text{False}, \text{True}), \text{True})\}$, bounded by $\uparrow((\text{True}, \text{True}), \text{True})$. This is the same defect `ScottHom.lean` records having already hit for the function space. The repair branches on the coordinatewise supremum landing in `NonBotPair` — the condition under which it is an element of $D \otimes E$ at all, rather than a merely sufficient condition for it. `smashSup_of_directed` and `smashSup_of_empty` kept their statements and were reproved, which is the agreement claim stated in Lean rather than in prose, and `smashCpo` needed no change.

Kernel check on the r0027 merges (`#print axioms`): all ten proved statements plus `smashCpo`, `smashSup_of_directed` and `smashSup_of_empty` depend only on `propext`, `Classical.choice`, `Quot.sound` — `lem10_prod` and `lem19` do not even need `Classical.choice`. None depends on `sorryAx`. `thm18` does, as its `sorry` requires.

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all twelve are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of $P \rightarrow N$] are just the finite subsets of N ”	p. 9	<code>isCompactElement_iff_finite</code>
2	“ $P \rightarrow N$... is a domain”	p. 9	<code>instance :</code> <code>Domain (Set X)</code>
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	<code>instance :</code> <code>CompletePartialOrder</code> <code>(ScottHom α β)</code>
4	“... bounded complete ... whenever E is”	Thm 7 proof	<code>instance :</code> <code>BoundedComplete</code> <code>(ScottHom α β)</code>
5	“step(s) ... is continuous”	Thm 7 proof	<code>scottContinuous_stepFun</code>
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	<code>isCompactElement_step</code>
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	<code>instance :</code> <code>IsAlgebraic</code> <code>(ScottHom α β)</code>
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	<code>exists_finite_isLUB_of_isCo</code>
9	“an embedding is an injection”	§3.1	<code>IsEmbeddingProjectionPair.i</code>
10	“a projection is a surjection”	§3.1	<code>IsEmbeddingProjectionPair.s</code>
11	“it is easy to check that p_N ... is a finitary projection”	§3.1	<code>isFinitaryProjection_normal</code>
12	“the set of strict continuous functions $D \rightarrow E$ is also a cpo”	§2.1	<code>ScottDomains.strictHomCpo</code>

Six of those eleven are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining theorems are supporting API: the $\ll$ calculus, the `compactsBelow` machinery, the pointwise order and suprema on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

Reference audit (r0020). Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for $\{\perp\} \in P(C)$, `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

The development is **33 modules, 8212 lines, 1 sorry, 0 other warnings**. Counts of definitions, results and theorems are in the Progress table above — they are not repeated here, so that this section cannot drift out of step with it. What each round delivered:

#	Round	Module	Contents
1	r0003	ScottDomains/WayBelow.lean	way-below $\ll$ at [Preorder α], 7 theorems; $x \ll x \leftrightarrow \text{IsCompactElement } x$ holds by <code>Iff.rfl</code>
2	r0004	ScottDomains/Domain.lean	<code>IsAlgebraic</code> , <code>Domain</code> (with the paper’s countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>
3	r0005	ScottDomains/Powerset.lean	the paper’s P N (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set N)</code> — the nondegenerate witness
4	r0006–r0007	ScottDomains/ScottHom.lean	the continuous function space $D \rightarrow E$: <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when E is — Theorem 7’s first sentence in full
5	r0008	ScottDomains/StepFunction.lean	the single step function <code>step k e</code> : continuity (from k compact), the adjunction $\text{step } k \ e \leq f \leftrightarrow e \leq f \ k$, and compactness in $D \rightarrow E$ (from e compact)
6	r0009	ScottDomains/FunctionSpaceDomain.lean	Domain is algebraic — the paper’s “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic)
7	r0010	ScottDomains/CompactFunction.lean	every compact function is a finite join of step functions — the finiteness half of the basis claim
8	r0011	ScottDomains/FunctionSpaceCountable.lean	<code>Countable</code> (Table E) is <code>countable</code> , hence <code>Domain (ScottHom α β)</code> — Theorem 7 complete
9	r0012	ScottDomains/NormalSubposet.lean	the normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012–r0013	ScottDomains/Projection.lean	embedding–projection pairs, projections, the paper’s “an embedding is an injection, a projection is a surjection”; $\text{im}(p)$ is a cpo and finitary projections
11	r0014	ScottDomains/FinitaryProjection.lean	Lemma 5 — the compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ (needs only that p is a projection), and $\text{im}(p) \cap K(D) \triangleleft K(D)$
12	r0015	ScottDomains/NormalProjection.lean	$p_N = \bigsqcup \{y \in N \mid y \sqsubseteq x\}$: continuous, a projection, and $\text{im}(p_N) \cap K(D) = N$ — half of Theorem 6’s correspondence, plus order preservation both ways
13	r0016	ScottDomains/Theorem6.lean	Theorem 6 — p_N is finitary, $p_{\{\text{im}(p) \cap K(D)\}} = p$, and the correspondence assembled
14	r0017	ScottDomains/FixedPoint.lean	Theorem 1 — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous f on a cpo. Not Mathlib reuse; see the §2 table

#	Round	Module	Contents
15	r0018	ScottDomains/UniformFixedPoint	Theorem 3 — fix is the unique uniform fixed-point operator; $\downarrow a$ as a cpo
16	r0019	ScottDomains/Product.lean	$D \times E$ as a cpo (the one construction needing no case split) and Lemma 8 parts 1–3
17	r0021	ScottDomains/Currying.lean	Lemma 8.4 — currying, $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$, and with it Lemma 8 complete
18	r0022	ScottDomains/EffectivePresentation	§2.1 effective presentation — the enumeration of the basis with its two decidability conditions
19	r0023	ScottDomains/Lift.lean	the lift $D \downarrow$ as a cpo, on Mathlib’s WithBot
20	r0024	ScottDomains/StrictHom.lean	the strict function space $D \rightarrow_{\perp} E$ as a cpo — needs no case split, since both branches of ScottHom’s $s\text{Sup}$ are strict
21	r0025	ScottDomains/Smash.lean	§4.3’s smash product $D \otimes E$ — the non-bottom pairs with a new bottom adjoined, as a cpo
22	r0025	ScottDomains/Bifinite.lean	§6.1’s Plotkin order and bifinite domain

§2 and §3 are now complete — the only §3 omission is the paper’s *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

Dependency structure of the remaining definitions

#	Definition	Prerequisites	Status
1	smash product $D \otimes E$	Domain, OrderBot	✓ done, r0025
2	bifinite / Plotkin order	IsNormalIn ($\triangleleft$, r0012)	✓ done, r0025
3	sum $D + E$	Domain	independent, ready
4	D_{∞}	embedding–projection pairs (r0012)	independent, but large (inverse limits)
5	the three powerdomains	the ideal completion (Theorem 11) — <i>not</i> built	blocked ; the three are independent of each other once it exists

Rows 1 and 2 were written as two modules and checked in a single build, which is the parallelism the structure actually admits — the bottleneck is the build-and-fix loop, not the writing.

Next: the sum $D + E$, then Theorem 11 to unblock the powerdomains, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo, $\perp$, monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s OrderHom.lfp is Knaster–Tarski over a complete lattice, not Kleene’s $\bigsqcup_n f^n(\perp)$ over a cpo. It is proved in ScottDomains/FixedPoint.lean, so **29** numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (IsCompactElement — its *definition* is already stated at [PartialOrder α] and so applies to a dcpo verbatim; what is CompleteLattice-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (WithBot/Sum), ideal completion (Order.Ideal).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — ScottDomains/WayBelow.lean), algebraic cpo, **domain**, bounded-complete (**all three done** — ScottDomains/Domain.lean),

embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D^∞ .

- **Theorems to prove (28):** 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

Lean column legend: ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop fi ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\sqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	✓ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	✓ <code>DirectedOn</code> / <code>IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\sqcup M$	✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	✓ <code>OrderBot</code>
2.1	Def	—	monotone function	✓ <code>Monotone</code> / <code>OrderHom</code>
2.1	Def	—	continuous : monotone, $f(\sqcup M) = \sqcup f(M)$ for directed M	✓ <code>ScottContinuous</code>
2.1	Thm	1	Fixed-Point Theorem : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\sqcup_n f^n(\perp)$	✓ proved (r0017) — <code>ScottDomains.theorem1</code> . Not Mathlib reuse: <code>OrderHom.lfp</code> is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem
2.2	Thm	2	Schröder–Bernstein for sets	✓ <code>Function.schroeder_bernstein</code> (<code>SetTheory/Cardinal/SchroederBernstein.lean</code>) — the name in an earlier draft of this row, <code>Function.Embedding.schroederBernstein</code> , does not exist
2.3	Def	—	fixed-point operator (uniform)	✓ <code>OrderHom.lfp</code> / <code>LawfulFix</code>
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	✓ proved (r0018) — <code>ScottDomains.theorem3</code> ; a fixed point operator is formalized as a family over every <code>CompletePartialOrder</code> in a universe

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element x : $x \sqsubseteq \sqcup M$ (dir.) $\implies x \sqsubseteq y$ some $y \in M$; $K(D)$	~ <code>IsCompactElement</code> — def is <code>[PartialOrder]</code> , so reusable on a <code>dcpo</code> ; its lemmas are all <code>CompleteLattice</code> -only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✓ <code>ScottDomains.WayBelow</code> (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by <code>Iff.rfl</code>

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✓ <code>ScottDomains.IsAlgebraic</code> (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)	✓ <code>ScottDomains.Domain</code> (r0004)
3.1	Def	—	bounded complete : $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ <code>ScottDomains.BoundedComplete</code> (r0004); the compound is <code>[Domain α] [BoundedComplete α]</code>
3.1	Def	—	(countably based) algebraic lattice	✓ <code>IsCompactlyGenerated</code> (+ <code>CompleteLattice</code>)
3.1	Def	—	embedding–projection pair (g, f)	✓ <code>ScottHom.IsEmbeddingProjectionPair</code> (r0012)
3.1	Def	—	projection; finitary projection p : $p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✓ <code>ScottHom.IsProjection</code> (r0012), <code>ScottHom.IsFinitaryProjection</code> (r0013) — $\text{im}(p)$ carries a <code>CompletePartialOrder</code> via <code>IsProjection.rangeCompletePartialOrder</code>
3.1	Def	—	normal subposet / substructure	✓ <code>ScottDomains.IsNormalIn</code> , notation $\triangleleft$ (r0012)
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$, and $\text{im}(p) \cap K(D) \triangleleft K(D)$	✓ proved (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i>) and <code>IsFinitaryProjection.isNormalIn_compacts</code>
3.1	Thm	6	Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)	✓ proved (r0015–r0016) — <code>ScottDomains.theorem6</code>
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✓ proved (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; D bounded complete is not needed
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s computable function is <i>not</i> formalized: it needs “recursively enumerable”, and Mathlib v4.32.2 has no <code>RePred</code> or equivalent (<code>grep</code> finds none) — building it is recursion theory, not domain theory

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ <code>Prod</code> order from Mathlib; the cpo instance is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance
4.2	Def	—	Church’s λ-notation (continuous abstraction)	✓ <code>OrderHom / ωCPO ContinuousHom</code>

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.3	Def	—	smash product $D \otimes E$	✓ <code>ScottDomains.Smash + smashCpo</code> (r0025)
4.4	Def	—	sum $D + E$; lift $D \perp$	lift ✓ <code>ScottDomains.liftCpo</code> on <code>WithBot</code> (r0023) — Mathlib gives the order, not the cpo; sum still ~
4.x	Lem	8	$D \times E \cong E \times D$; $(D \times E) \times F \cong D \times (E \times F)$; $D \rightarrow (E \times F) \cong (D \rightarrow E) \times (D \rightarrow F)$; $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$	✓ proved — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ prove
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete	✗ prove
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	~ <code>Order.Ideal</code> exists $\rightarrow$ prove
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✗ prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ prove

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✗ define
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✗ define
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain	✓ <code>ScottDomains.IsPlotkinOrder, IsBifinite</code> (r0025)
6.1	Prop	15	Every bounded-complete domain is bifinite	✗ prove
6.1	Thm	16	D bifinite $\implies Fp(D)$ is an algebraic lattice	✗ prove
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \times, \otimes, +, () \perp$ bifinite (incl. function space)	✗ prove
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove
6.2	Lem	19	closure $r : D \rightarrow D$ ($r \circ r = r \sqcap id$) $\implies im(r)$ is a domain	✗ prove
6.2	Lem	20	D domain $\implies Fc(D)$ (finitary closures) is a cpo	✗ prove

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain / D^∞ (inverse limit)	X define
7	Thm	21	F representable over $\text{cpo } U \implies$ a domain D with $D \cong F(D)$	X prove
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	$\sim \text{IsCompactlyGenerated} \rightarrow$ prove
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	X prove
7	Lem	24	$U \text{ cpo}; \times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)	X prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies$ U universal	X prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	X prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	X prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ representable over U	X prove
7	Thm	29	D bifinite $\implies D^+$ bifinite; solving $D \cong D^+$	X prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ p-representable over universal bifinite V	X prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · $\times$ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D^∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.